

## **PROJECT-1**

**Pandas** is a powerful Python library used for data analysis and manipulation. It helps you work with structured data like tables (similar to Excel or SQL tables).

Here are the most important Pandas file-import methods in data analysis. Pandas provides simple functions to read different file format into a Data Frame.

The dataset Importing a CSV file (railway\_gauges 1.csv) is loaded into a DataFrame using `pd.read_csv()`.

```
#Load csv file
df=pd.read_csv("railway_gauges 1.csv")
print(df.head())
```

**Figure-1 :Loading CSV file into**

**pandas**

Using Pandas data Frame object we can see **head()**,**tail()**functions to display top 5,last 5 rows from imported dataset.

```
===== RESTART: C:/Users/LENOVO/Desktop/Analytics/sample_Project/main.py =====
   Year  Broad Gauge  Metre Gauge  Narrow Gauge  Total
0  1964-65         3521         2891         464   6876
1  1965-66         3603         2913         470   6986
2  1966-67         3615         2911         481   7007
3  1967-68         3601         2920         506   7027
4  1968-69         3603         2921         508   7032
```

**Figure-2: Displaying top 5 rows**

Some of the Pandas functions are:

**read\_csv()**-Loads CSV file into a Data Frame

**iloc[]** - used for selecting data by integer position

**idxmax()**-Returns index of maximum value

```
# find which year had the maximum installations
max_row=df.iloc[[df['Total'].idxmax()]]
print(max_row)
```

|    | Year    | Broad Gauge | Metre Gauge | Narrow Gauge | Total |
|----|---------|-------------|-------------|--------------|-------|
| 48 | 2012-13 | 6577        | 379         | 216          | 7172  |

**Figure-3 : Displaying details of highest total Year**

I used pandas functions such as `read_csv()` to load the dataset, `head()` to preview data, `fillna()` to handle missing values, and `to_numeric()` for type conversion. I also used `sum()` for aggregation and `loc[]` for filtering. Attributes like `columns` and `dtypes` were used to understand the dataset structure. Additionally, I performed column selection, filtering, and feature creation for analysis.

**Matplotlib** is a popular Python library used for data visualization. It allows you to create charts and graphs to better understand and present data.

It works well with libraries like NumPy and pandas.

Some of the Matplotlib functions are:

**plt.plot()** - is used to create a line graph.

**plt.bar()** - creates a bar chart.

**plt.show()** - Displays the plot.

## DATA VISUALIZATION

Matplotlib is used to visualize data in graphical form, helping to analyze trends, compare values, and understand large datasets easily. It is

widely used in data science, machine learning, and scientific computing for creating charts like line graphs, bar graphs, and pie charts.

This code is used to visualize the relationship between Year and Number of railway tracks using a bar graph with Matplotlib.

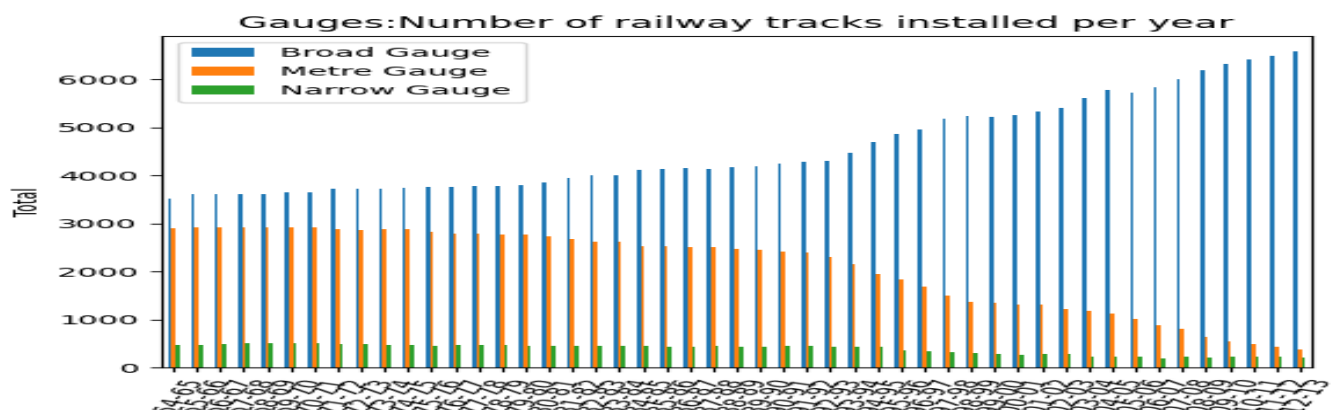
```
#Plot data using bar chart
df=df.drop('Total',axis=1)
ax=df.plot(x="Year",kind="bar")
plt.xticks(rotation=70)
plt.xlabel('Year')
plt.ylabel('Total')
plt.title('Gauges: Number of railway tracks installed per year')
plt.savefig('rail_gauges.png')
plt.show()
```

**Figure-4 : Bar Graph Code**

The resulting graph shows how the number of railway tracks changes over different years, helping to identify trends such as increase or decrease over time.

## RESULTS

The project successfully analyzes railway gauge data using pandas and Matplotlib, and concludes that Broad Gauge dominates the modern railway system with overall increasing track growth.



**Figure-5 : Number of Railway Tracks installed per  
year**